



GS FOUNDATION 2024-25

GEOGRAPHY - 3

Composition and Structure of the Atmosphere

Insolation, Heat Budget and Temperature

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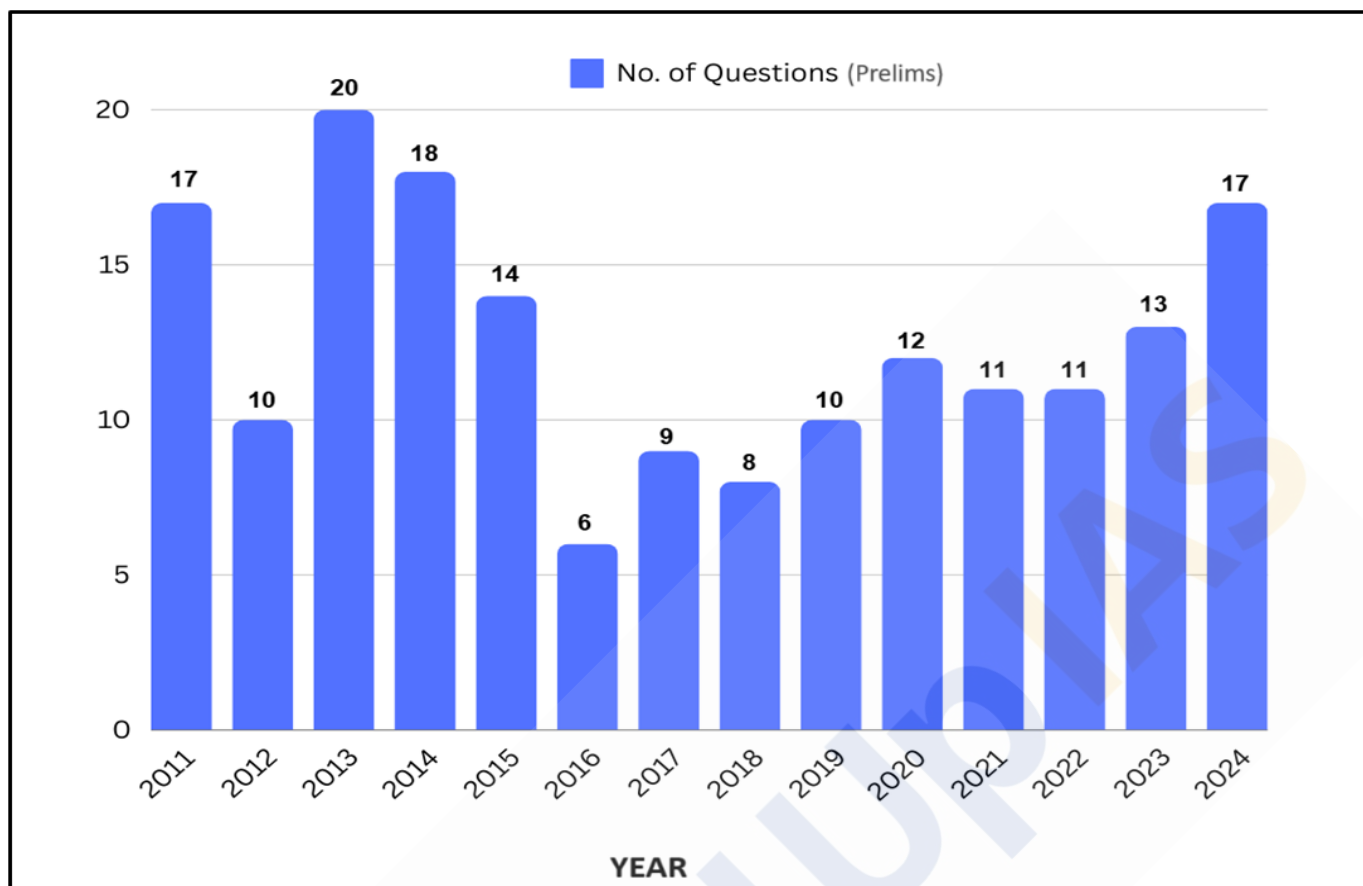
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Composition and Structure of the Atmosphere

Introduction:

The atmosphere is the envelope of gases surrounding Earth, forming a **protective boundary** between outer space and the biosphere. It consists of a **dynamic collection of gases** that constantly move and change. These gases are organized into layers defined by composition and temperature.

Atmosphere is held to the Earth by the **force of gravity**, and as a result, most of what comprises the atmosphere is towards the Earth's surface.

- **50%** of the atmospheric mass contained within the first **5 – 6 km** from the Earth's surface.
- **90%** of the atmospheric mass contained within the first **30 km** from the Earth's surface.
- **99%** of the atmospheric mass contained within the first **40 km** from the Earth's surface.

Though the atmosphere can be traced up to a height of 10,000 km, for practical consideration we generally take the height of the atmosphere up to 450 km from the Earth's surface.

The gases of the present atmosphere are **not the direct residue of the early stage of earth's formation**. They are a product of progress through **volcanic eruptions, hot springs, chemical breakdowns** of solid matter and redistribution from the biosphere.

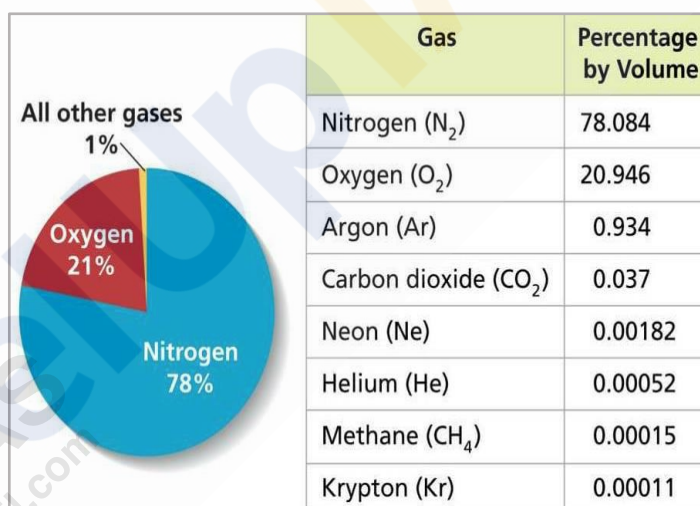
Roles of Earth's Atmosphere:

- **Protection and Shielding** - The atmosphere serves as a protective shield for Earth, shielding it from harmful solar radiation such as ultraviolet (UV) rays. **Ozone (O₃)** in the stratosphere absorbs most of the **Sun's UV radiation**, preventing it from reaching the Earth's surface where it could cause damage to living organisms and DNA. This protective function is crucial for maintaining life on Earth's surface.
- **Regulation of Temperature** - Through the **greenhouse effect**, the atmosphere regulates Earth's temperature. It **allows incoming shortwave solar radiation** to penetrate the atmosphere and reach the Earth's surface. The surface absorbs this energy and re-emits it as longer-wavelength infrared radiation. Greenhouse gases like carbon dioxide (CO₂), methane (CH₄), and water vapour (H₂O) trap some of this outgoing infrared radiation, preventing it from escaping directly into space. This process keeps the Earth's surface warmer than it would be without an atmosphere, making it habitable for life.
- **Distribution of Solar Energy** - The atmosphere plays a crucial role in distributing solar energy across the Earth's surface. It redistributes heat **from the equator to the poles through atmospheric circulation** patterns such as Hadley cells, Ferrel cells, and polar cells. These circulation patterns, driven by solar heating and Earth's rotation, transport heat and moisture, influencing global climate patterns and weather phenomena.
- **Facilitation of Weather Phenomena** - Weather patterns and phenomena such as **clouds, precipitation, and storms** are facilitated by the atmosphere's dynamics. Temperature differences between land and sea, variations in solar heating, and the Earth's rotation create atmospheric instability that leads to the formation of weather systems. Atmospheric layers like the troposphere, where most weather occurs, and the stratosphere, which contains the ozone layer, play critical roles in shaping weather and climate.

- **Atmospheric Pressure and Circulation** - Atmospheric pressure, exerted by the weight of air above, influences weather patterns and circulation. Variations in pressure drive winds and ocean currents, which in turn distribute heat and moisture around the globe. The interaction between high-pressure and low-pressure systems results in the formation of wind patterns such as trade winds, westerlies, and monsoons, influencing regional climates and weather events.
- **Protection from Meteoroids** - The atmosphere provides protection from meteoroids and **space debris** that enter Earth's atmosphere. As these objects enter the atmosphere at high speeds, they experience frictional heating due to air resistance. This heat causes them to **burn up and disintegrate before reaching the Earth's surface**, reducing the impact of potential collisions.
- **Support for Life and Biospheric Processes** - Life on Earth depends on the atmosphere for **oxygen (O₂)** and **carbon dioxide (CO₂)** exchange essential for **respiration and photosynthesis**. Atmospheric gases like **nitrogen (N₂)** and **oxygen (O₂)** are critical for **supporting biological processes** and sustaining ecosystems. The atmosphere also moderates temperature extremes between day and night, creating stable conditions necessary for life to thrive.

Composition of the Atmosphere:

The Earth's atmosphere is a complex **mixture of various gases, water vapour, and solid particles**. The composition of the atmosphere is not static and it **changes according to the time and place**. This blend of components plays a crucial role in supporting life, influencing climate, and facilitating various geophysical processes.



Major Components of the Atmosphere:

Nitrogen (N₂):

- Nitrogen is the **most abundant gas**, making up about **78%** of the atmosphere. It is a relatively **inert gas** and crucial for living organisms, as it forms a **fundamental part of amino acids, proteins, and nucleic acids**. It dilutes the activity of oxygen and helps in the combustion of fuels and food at moderate rate. If nitrogen were not there, burning would be so fast and violent that any fire would be very difficult to control. It is removed from the atmosphere by **soil bacteria** through **nitrogen fixation**, converting it into forms usable by plants. When plants and animals decompose, decomposer bacteria break down their organic matter, releasing nitrogen back into the atmosphere, thus maintaining the nitrogen cycle.

Oxygen (O₂):

- Despite making up only **21%** of the atmosphere's total volume, oxygen is the most important gas. **All aerobic organisms require oxygen for respiration**. Additionally, oxygen is **essential for combustion** and combines with other elements to form oxides. This gas plays a significant role in **maintaining the Earth's biosphere and energy cycles**. Oxygen is removed from the atmosphere when organic matter decays, when it combines with other substances, or when it is consumed during breathing. It is added back to the atmosphere through photosynthesis by plants.

Argon (Ar):

- Argon makes up roughly **0.93%** of the atmosphere. As an **inert gas** with limited interaction with other elements, it contributes to the stable makeup of the atmosphere. Released from **Earth's crust** and mantle through volcanic degassing, Argon is used in various industrial applications due to its inert properties.

Carbon dioxide (CO₂):

- Carbon dioxide, though only about **0.038%** of the dry atmosphere by volume, is a critical **greenhouse gas** that absorbs heat and plays a significant role in the **Earth's heat energy budget**. It is essential for photosynthesis in plants and has been increasing due to the burning of fossil fuels, contributing to global warming.

Ozone (O₃):

- Ozone is a **trace gas** in the atmosphere, constituting less than **0.00003%** by volume. It is concentrated mainly between 20 to 25 kilometers altitude (stratosphere), where it forms the **ozone layer**. This layer is vital for life on Earth as it absorbs the majority of the **Sun's harmful ultraviolet radiation**. Ozone's presence is essential for protecting living organisms from UV-induced damage. Very little is found naturally near the ground where it is a toxic pollutant. Sometimes dangerously high concentrations develop near large cities in a process called **photochemical smog**.

Other Gases:

- In addition to the major components, the atmosphere contains several trace gases, including **Neon (Ne)**, **Helium (He)**, **Hydrogen (H₂)**, **Xenon (Xe)**, **Krypton (Kr)**, and **Methane (CH₄)**. These gases, although present in minute quantities, play various roles in atmospheric processes and industrial applications. Methane, in particular, is a potent greenhouse gas produced by natural processes and human activities.

Water Vapour:

- Water vapour is also a **gas** in the atmosphere, which **decreases with altitude**. It is one of the **most variable components** of the atmosphere, constituting between **0 to 4% by volume**, depending on the location and temperature. 90% of moisture content in the atmosphere exists within 6 km of the surface of the earth. It is primarily introduced into the atmosphere through the **evaporation** of water from oceans, lakes, rivers, and vegetation. Water vapour plays a **critical role in weather and climate** by participating in the hydrological cycle and influencing the greenhouse effect. It **absorbs both incoming solar radiation and outgoing terrestrial radiation**, thus regulating temperature. It is **essential for the formation of clouds and precipitation**, and its **capacity to hold moisture is directly proportional to temperature**. In the tropics, water vapour can account for up to 4% of the air by volume, while in polar regions, it can be less than 1%.

Particulate Matter (Aerosols):

- The Solid Particles present in the atmosphere consist of **sand particles** (from weathered rocks and also derived from volcanic ash), **pollen grains**, **small organisms**, **soot**, **ocean salts**. These particulates help in the **absorbing, reflecting, and scattering of the solar radiation** which adds the varied charming colour of red and orange at sunrise and sunset. The sky appears blue in colour due to the selective scattering of solar radiation by dust particles. Salt particles become **hygroscopic nuclei** and thus help in the **formation of water drops, clouds, and various forms of condensation and precipitation**.

The atmosphere's composition is a delicate balance of gases, water vapour, and particulate matter, each contributing uniquely to the Earth's climatic and life-supporting systems. Understanding these components is essential for grasping the complexities of weather, climate change, and environmental interactions.

Thermal Zonation:

Tropo



Temp. Difference is seen

Troposphere:

- The troposphere is the **lowest layer of Earth's atmosphere**, extending from the surface up to an altitude of about **8 km** at the **poles**, **13 km** at mid-latitudes, and **18 km** at the **equator**. This variation in thickness is due to the heating of air at the equator, causing it to rise to greater heights compared to the poles.
- It contains approximately **75-80% of the atmosphere's mass**, making it the **densest** of all atmospheric layers.
- One of the key characteristics of the troposphere is the decrease in temperature with altitude. On average, the temperature drops by approximately **6.5 °C per kilometre**, known as the "Lapse rate."
 - At the **tropopause**, the boundary between the troposphere and the stratosphere, temperatures can reach as low as **-45°C over the poles** and **-80 °C over the equator**. The greater fall in temperature above the **equator** is due to the greater thickness (height) of the troposphere in that region. *Temp. Diff due to height.*
- It is the most meteorologically significant part of the atmosphere, where **nearly all-weather phenomena occur**, including rainfall, fog, hailstorms, cyclones, and anticyclones. This is due to its high concentration of atmospheric water vapour and solid particles essential for cloud formation and precipitation.
- The troposphere, often called the "convective region," is where vertical convective currents occur due to the heating of the Earth's surface. **Warm air rises while cool air descends**, driving weather formation and energy transfer. The **convective activity stops at the tropopause**. The tropopause is characterized by relatively **constant temperatures**, and it marks the limit beyond which convection cannot occur.

Stratosphere:

- The stratosphere is the **second layer of Earth's atmosphere**, situated **above the troposphere** and extends up to about **50 kilometers** above the Earth's surface. Its upper boundary is the **stratopause**.
- A key feature of the stratosphere is the **Ozone layer**, which contains a high concentration of ozone (**O₃**) molecules. This layer is critical for life on Earth as it **absorbs the majority of the Sun's harmful ultraviolet radiation**, preventing it from reaching the Earth's surface where it can cause skin cancer, cataracts, and other health issues, as well as damaging crops and marine ecosystems.
- Unlike the troposphere, the **temperature in the stratosphere** initially remains constant for a short distance and then gradually **increases with altitude**. At around 50 kilometers, the temperature reaches approximately 0°C. This increase in temperature with height is primarily due to the **absorption of UV radiation by ozone molecules**, which release heat in the process.
- It is almost **free from weather-related phenomena** such as clouds, making it an **ideal zone for commercial aviation**. Aircraft often fly in the lower stratosphere or upper troposphere where conditions are calm and stable.
 - At high latitudes during winter, **Polar Stratospheric Clouds** can form in the **coldest altitudes** of the stratosphere. These clouds play a role in chemical reactions leading to the depletion of ozone, especially over polar regions.

Mesosphere:

- The mesosphere is an intermediate atmospheric layer situated **above the stratosphere**, extending from approximately **50 to 80 kilometers** above Earth's surface. Its upper boundary is the **mesopause**.

- In the mesosphere, **temperatures generally decrease with altitude**. The top of the mesosphere, known as the **mesopause**, can see temperatures dropping to around -100°C (-120°F). This temperature can vary, but the average at the mesopause is about -85°C . This makes the mesopause one of the **coldest places in Earth's atmosphere**, especially during the summer near the poles.
- One of the defining features of the mesosphere is its role in **burning up meteorites** that enter the Earth's atmosphere from space. Most meteors disintegrate in this layer due to the intense friction and heat generated upon entry, creating a spectacular display of shooting stars.
- The very scarce water vapour present at the top of the mesosphere can form **'Noctilucent clouds,'** which are the **highest clouds in Earth's atmosphere**. These clouds are visible during twilight and are known for their **striking luminescence**.

Thermosphere:

- The thermosphere is located above the mesopause, spanning from approximately **80 to 400 kilometers** above Earth's surface.
- It is characterized by a significant **rise in temperature with increasing altitude**, reaching up to $2,000^{\circ}\text{C}$ ($3,600^{\circ}\text{F}$). This temperature increase is primarily due to the **absorption of energetic ultraviolet (UV) and X-ray radiation from the sun**. Despite these high temperatures, a person or object in the thermosphere would not feel the heat due to the extremely low pressure and sparse distribution of gas molecules, which are spaced hundreds of kilometers apart.
- The **ionosphere**, part of the thermosphere, extends from 80 to 400 km. It **reflects radio waves**, enabling long-distance communication.
- Phenomena such as the **aurora borealis** (Northern Lights) and **aurora australis** (Southern Lights) are often observed in the lower parts of the thermosphere. These stunning light displays are the result of **interactions between solar winds and the Earth's magnetic field**.
- **The International Space Station (ISS)** and various satellites orbit within the thermosphere. This layer provides an ideal environment for such orbits due to its low density, which reduces drag on spacecraft.

Exosphere:

- The exosphere is the outermost layer of Earth's atmosphere, extending **beyond the thermosphere above a height of about 400 kilometers**. It is situated between approximately **700 and 10,000 kilometers** above the Earth's surface, where it gradually merges with the solar wind and outer space.
- This layer is composed predominantly of **light gases such as hydrogen and helium**. Oxygen atoms are also present but in very low density.
- **The temperature in the exosphere increases with altitude** as it is directly exposed to solar radiation. This is because there are very few gas molecules to absorb and reradiate the heat.
- The exosphere essentially marks the **boundary of Earth's atmosphere and outer space**. It coincides with space and blends into the vacuum of outer space.
- Most **Earth-orbiting satellites** operate within the exosphere due to its minimal **atmospheric drag**, allowing them to maintain their orbits for extended periods.

Compositional Zonation:

Homosphere:

- The homosphere is **the lower segment of the two-part division of atmosphere** and further consists of three regions namely **troposphere, stratosphere and mesosphere**. It extends from the earth's surface up to an altitude of **80-100 km**.
- Troposphere is the bottom-most layer of the homosphere. The Stratosphere is the middle region of the Homosphere. The Mesosphere is the top layer of the Homosphere.
- The bulk **composition of the air** (gases) is mostly **uniform** so the **concentrations of molecules are the same** throughout the homosphere. However, the concentration of air keeps decreasing significantly as the altitude increases.
- The top of the homosphere is called the **homopause**, also known as the **turbopause**.

Heterosphere:

- Above the homopause is the heterosphere, encompassing the region from **80 km to about 10,000 km**, this layer includes both the ionosphere and exosphere.
- The **air** (gases) is **no longer uniform** as diffusion is faster than mixing. **The heavy gases decrease in density with altitude more rapidly than lighter gases**. This separation of gases is caused by Earth's **gravity** in which **heavier gases are pulled closer to the surface** and the **lighter gases drift farther outward**.

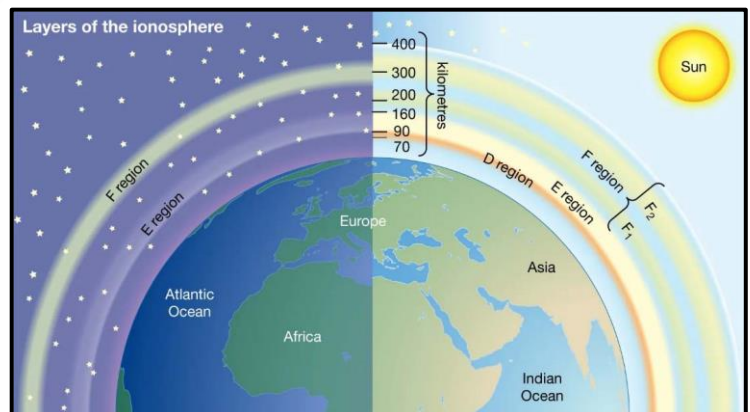
Functional Zonation:

Ozonosphere:

- It lies at an altitude **between 30 km and 60 km** from the Earth's surface and spans the stratosphere and lower mesosphere.
- This layer is crucial as it absorbs most of the sun's **harmful ultraviolet (UV) radiation**, particularly **UV-B** and **UV-C** rays, thus protecting living organisms from potential damage such as skin cancer, cataracts, and immune system harm.
- The ozonosphere is also called **chemosphere** because, a lot of chemical activity goes on here. The temperature rises at a rate of **5°C per Kilometer** through the ozonosphere.

Ionosphere:

- Unlike the distinct layers mentioned earlier, the ionosphere is not a separate layer but rather **overlaps with the mesosphere, thermosphere, and exosphere**. It's a **very active** part of the atmosphere, and it **grows and shrinks** depending on the energy it absorbs from the sun.
- It is formed primarily by the **ionization of atmospheric gases due to solar and cosmic radiation** and is divided into **several layers - D, E, and F**. These layers vary in ionization levels and **influence on radio wave propagation**. The ionosphere **reflects radio waves**, enabling **long-distance communication** by bouncing signals between the ionosphere and Earth's surface. **Solar activity and geomagnetic storms can disrupt these signals**, impacting communication and navigation systems.



Insolation, Heat Budget and Temperature

Introduction:

The Earth maintains a stable overall temperature through a balanced exchange of heat, ensuring it neither gains nor loses heat as a whole. This equilibrium is achieved through the balance between the amount of solar energy absorbed by the Earth, known as insolation, and the amount of energy radiated back into space, called terrestrial radiation. When the energy received from the sun equals the energy emitted by the Earth, a state of thermal equilibrium is reached. This balance is crucial for maintaining the Earth's climate system and supporting life. Any significant imbalance in this exchange can lead to global temperature changes, influencing weather patterns, climate stability, and overall environmental conditions.

Insolation or Incoming Solar Radiation:

The Earth's climate and weather systems are fundamentally driven by the **energy received from the Sun**, known as **insolation**. This term, derived from '**incoming solar radiation**,' refers to the solar energy that reaches the Earth's surface.

The Sun is the primary source of energy for Earth, radiating immense amounts of energy into space. However, only a fraction—**approximately two billionths**—of this solar radiation reaches the Earth's surface due to factors such as the Earth's size and distance from the Sun. This small proportion of solar radiation, while seemingly minute, plays a pivotal role in driving Earth's climate and ecosystems.

Absorption of energy on earth surface.

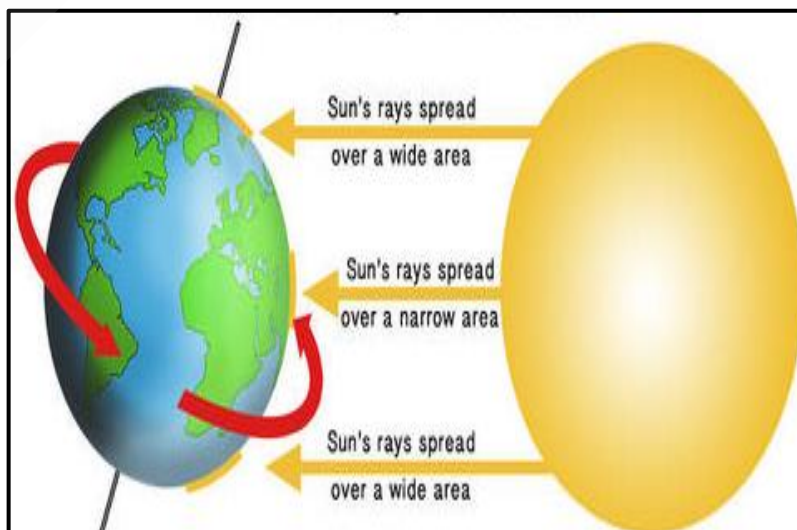
Factors affecting distribution of Insolation on Earth:

Rotation of Earth on its Axis:

- As the Earth rotates, **half of its surface is illuminated by the Sun, experiencing daylight and receiving direct solar radiation**. Simultaneously, the **other half of the Earth is in shadow, experiencing darkness** and no direct solar insolation during that period. This daily cycle of rotation leads to significant variations in insolation between day and night, impacting temperatures, weather patterns, and biological rhythms across different regions.

Angle of Incidence of Sun's Rays:

- The **angle** at which solar rays strike the Earth's surface **varies with latitude**. Near the **equator, sunlight strikes more vertically**, concentrating its energy over a smaller area. This results in **higher insolation** per unit area. Conversely, at **higher latitudes, the angle of incidence becomes shallower**, spreading solar energy over a larger surface area. This phenomenon **reduces the intensity** of insolation received per unit area, contributing to cooler climates in these regions.



Length of the Day:

Longer day more sunlight, more insolation received

- Varying lengths of day and night, influenced by the Earth's tilt and its revolution around the Sun, affect the amount of insolation received. **The longer the day, the greater the amount of insolation received. Conversely, a shorter day leads to the receipt of less insolation.**

Transparency of the Atmosphere:

Clear sky more absorption

- The atmosphere plays a crucial role in modulating insolation by either transmitting, absorbing, or reflecting solar radiation. **Clear skies allow maximum solar energy to reach the Earth's surface, whereas clouds and atmospheric particles such as dust and water vapour can scatter and absorb incoming radiation.** Aerosols and pollutants in the atmosphere can also influence insolation distribution by altering the optical properties of the atmosphere, affecting how much sunlight reaches the surface.

Distance between the Earth and the Sun:

- The **distance between Earth and the Sun varies slightly due to its elliptical orbit**, affecting the total amount of solar radiation received (insolation). However, this variation has **minimal direct impact** on daily or seasonal weather patterns compared to angle of sun's rays and length of the day.

Sunspots:

- Sunspots are **dark areas on the Sun's photosphere** surrounded by the chromosphere. They form due to periodic solar disturbances and explosions, appearing cooler by about 500°C compared to their surroundings. The number of sunspots **varies cyclically over an 11-year period**. Sunspot activity influences the amount of radiation reaching Earth; higher sunspot numbers correlate with increased solar energy and, consequently, higher insolation levels on Earth's surface.

Geographical Features and Surface Characteristic:

- The configuration of land masses, oceanic expanses, and surface albedo (reflectivity) also impact insolation distribution. Surfaces with **higher albedo, such as snow-covered regions, reflect more sunlight back into space, reducing local heating.** Conversely, **darker surfaces like forests and oceans absorb more solar radiation**, leading to higher temperatures.

more reflection back to atmosphere

Albedo

Albedo is a measure of the **reflectivity of a surface**. It is defined as the ratio of the amount of solar radiation reflected by a surface to the amount of solar radiation received by that surface.

Albedo is influenced by several factors:

- Surface Type and Composition** - Smooth surface have higher reflectivity.
- Angle of Incoming Solar Radiation** - Higher reflectivity in case of low angle.
- Atmospheric Conditions** - Clouds have high albedos.
- Seasonal Changes** - Snow cover in winter increases albedo.

Surface type	Typical value
Fresh asphalt	0.03 – 0.04
Open ocean	0.06
Conifer forest (summer)	0.08 – 0.15
Worn asphalt	0.12
Deciduous trees	0.15 – 0.18
Sand	0.15 – 0.45
Tundra	0.18 – 0.25
Agricultural crops	0.18 – 0.25
Bare soil	0.17
Green grass	0.20 – 0.25
Desert sand	0.30 – 0.40
Snow	0.40 – 0.90
Ocean ice	0.50 – 0.70
Fresh snow	0.80 – 0.90

more reflective

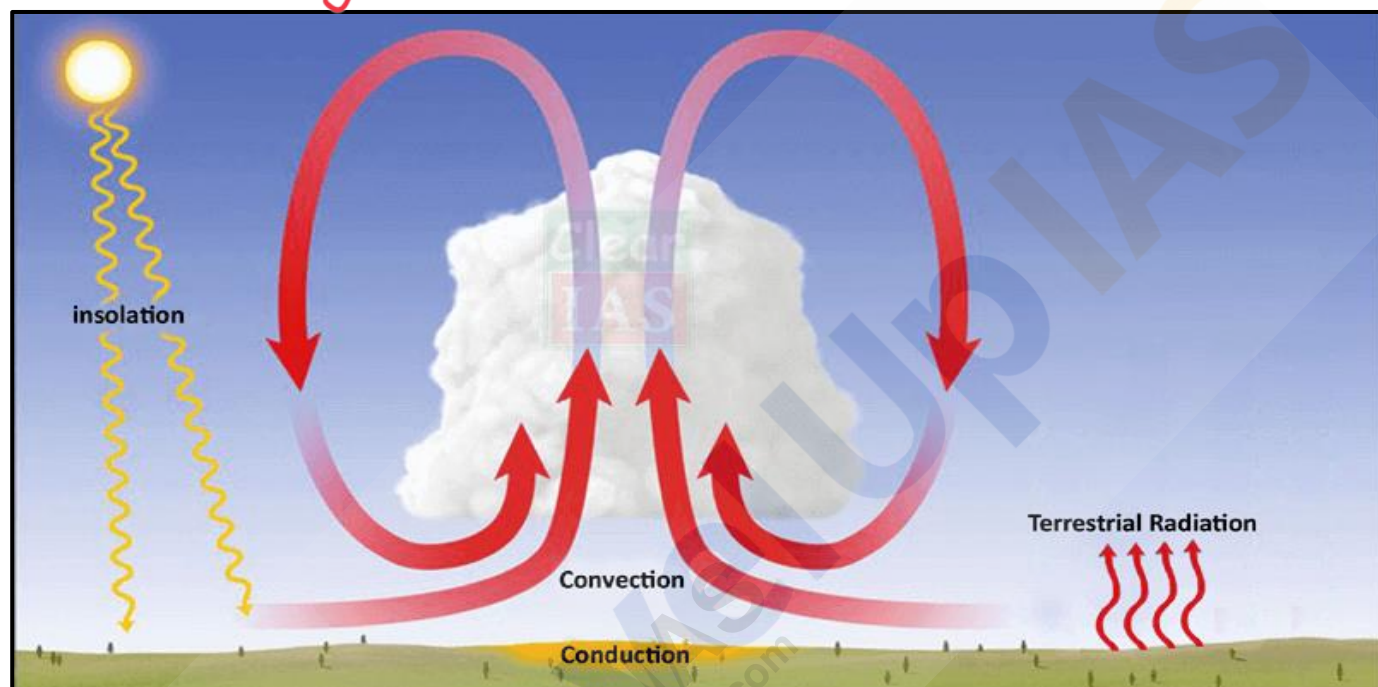
Heating and Cooling of the Atmosphere:

The Sun is the primary source of atmospheric heat and energy, but its effect is indirect. For instance, when we climb a mountain or ascend in the atmosphere, the temperature decreases rather than increases, contrary to what we might expect. This is due to the complex mechanism of heating the atmosphere.

Four processes are directly responsible for this heating:

- Conduction
- Convection
- Advection
- Terrestrial Radiation

Why is there this behavior
 Agar upar jarcha, means going near sun
 means jada garam, but we see just ultain
 Troposphere



Conduction: → Transfer of heater b/w warmer & cooler object.

- When **two objects of unequal temperature** come in contact with each other, heat energy flow from the warmer object to the cooler object and this process of heat transfer is known as conduction. The flow continues till temperature of both the objects becomes equal or the contact is broken.
- The conduction in the atmosphere occurs at zone of contact between the atmosphere and the earth's surface. However, this is a minor method of heat transfer in terms of warming the atmosphere since it only affects the air close to the earth's surface.

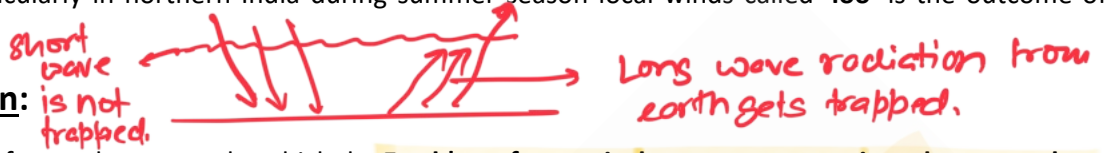
Convection: (Circular Motion) ↓ density > 1.29 kg/m^3 ∴ mass volume ↑

- Transfer of heat by movement of a mass or substance from one place to another, generally vertical, is called convection.
- The air of the lower layers of the atmosphere gets heated either by the earth's radiation or by conduction. The heating of the air leads to its expansion. Its density decreases and it moves upwards.
- Continuous ascent of heated air creates vacuum in the lower layers of the atmosphere. As a consequence, cooler air comes down to fill the vacuum, leading to convection. The cyclic movement associated with the convective process in the atmosphere transfer heat from the lower layer to the upper layer and heats up the atmosphere.

Advection: { Horizontal movement of wind }

- It is the transfer of heat through horizontal movement of winds. Winds carry the temperature of one place to another. The temperature of a place will **rise** if it lies on the **path of winds coming from warmer regions**. The temperature will **fall** if the place lies on the **path of the winds blowing from cold regions**. This process of horizontal transport of heat by winds is known as **advection**.
- In middle latitudes, most of diurnal (day and night) variation in daily weather are caused by advection alone. In tropical regions particularly in northern India during summer season local winds called 'loo' is the outcome of advection process.

Terrestrial Radiation:



- Terrestrial radiation refers to the process by which the Earth's surface emits long-wave energy into the atmosphere after absorbing incoming solar radiation (insolation). This long-wave energy heats the atmosphere from the surface upwards.
- The atmosphere is transparent to incoming short-wave solar radiation but absorbs outgoing long-wave terrestrial radiation, particularly by **greenhouse gases** like carbon dioxide. This absorption contributes to the greenhouse effect, trapping some heat and maintaining Earth's average temperature suitable for life.

Heat Budget of the Earth:

The heat budget of the Earth refers to the **balance between the incoming solar radiation (insolation) and the outgoing terrestrial radiation**. This balance is crucial for maintaining the Earth's temperature and climate stability.

Insolation and Albedo:

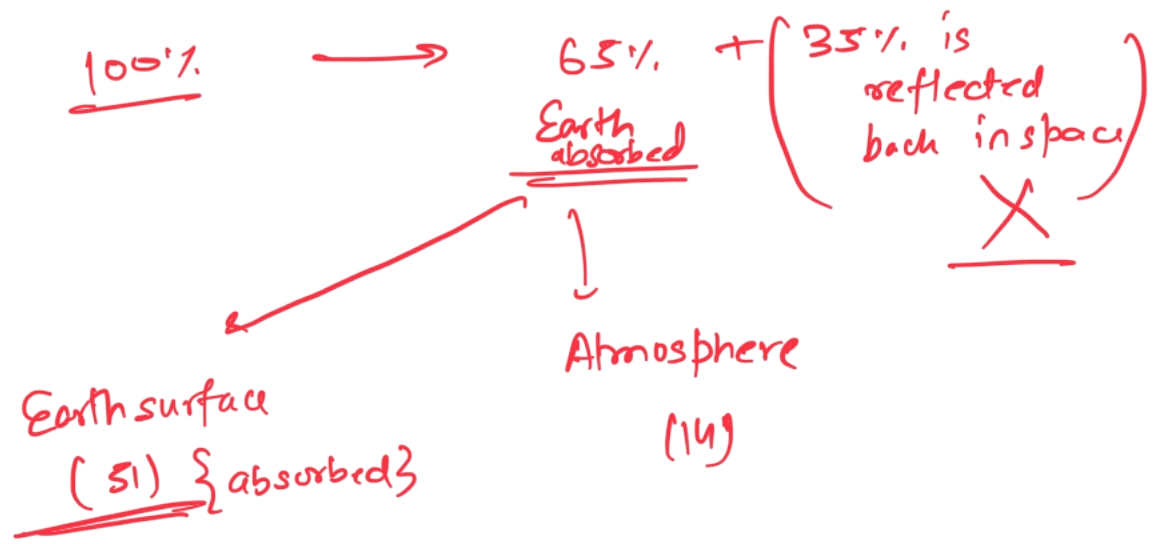
- Insolation:** 100% of Solar radiation from the Sun reaches the Earth's atmosphere.
- Albedo:** Approximately 35% of the incoming solar radiation is immediately reflected back into space before reaching the Earth's surface. This reflected amount is known as albedo.
 - Reflected from the top of the atmosphere - 6 units.
 - Reflected by clouds - 27 units.
 - Reflected from snow and ice - 2 units.

Absorption and Distribution of Solar Energy:

- Surface Absorption:** Of the remaining **65%** of solar radiation reaching the Earth's surface, **51% is absorbed by the Earth's surface**.
 - This includes **34%** through **direct solar radiation** and **17%** from **scattered radiation**.
- Atmospheric Absorption:** The atmosphere absorbs about **14%** of incoming solar radiation.

Terrestrial Radiation:

- Surface Radiation:** The Earth's surface radiates back approximately **51%** of the absorbed solar energy as **terrestrial radiation**.
 - 34%** is **absorbed by the atmosphere**.
 - 17%** of this radiation is **emitted directly to space**.
- Atmospheric Radiation:** The atmosphere, in turn, radiates back **48% of the absorbed energy** to space.
 - 14%** from **incoming solar radiation**.
 - 34%** from **terrestrial radiation** (including the absorbed 17% directly emitted from the Earth's surface).



Then Earth radiate back some of the heat (51)

17% direct to space

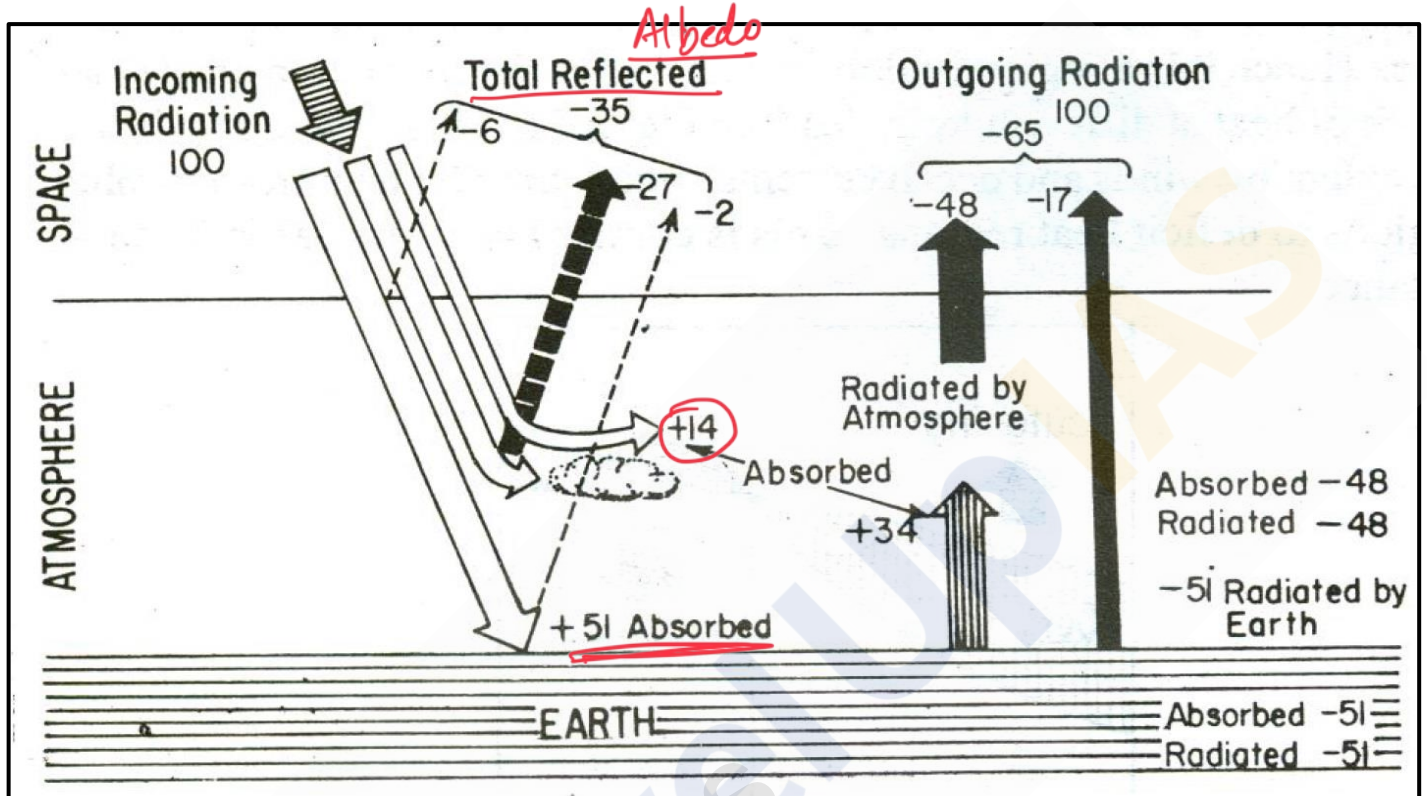
34% reflected by Atmosphere

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graph TD; G["Earth radiate back some of the heat (51)"] --> H["17% direct to space"]; G --> I["34% reflected by Atmosphere"];
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$\therefore$ Total reflected

Heat Budget (Heat Balance):

- The total radiation returning from the Earth's surface and the atmosphere combined is 17 (directly from the Earth) - 48 (from the atmosphere) = 65%.
- This **total radiation balance (65%)** is equal to the total of 65% received from the Sun (Solar Insolation). This equilibrium in heat intake and release ensures that the **Earth's temperature remains stable**, preventing it from either warming up or cooling down excessively.



Variation in the Net Heat Budget at the Earth's Surface:

Imbalance of Heat Distribution:

Despite the Earth maintaining an overall balance between incoming solar radiation and outgoing terrestrial radiation, significant variations exist across latitudes. This imbalance is primarily due to the **variation in the amount of insolation received**, which is directly related to latitude.

- Tropical Regions:** In tropical regions, located near the equator, the amount of **insolation received exceeds the amount of terrestrial radiation emitted**. This surplus results in these regions being characterized by excess heat.
- Polar Regions:** Conversely, at the poles, the amount of heat gained from **insolation is less than the heat lost through terrestrial radiation**. This deficit leads to polar regions being regions of heat loss.

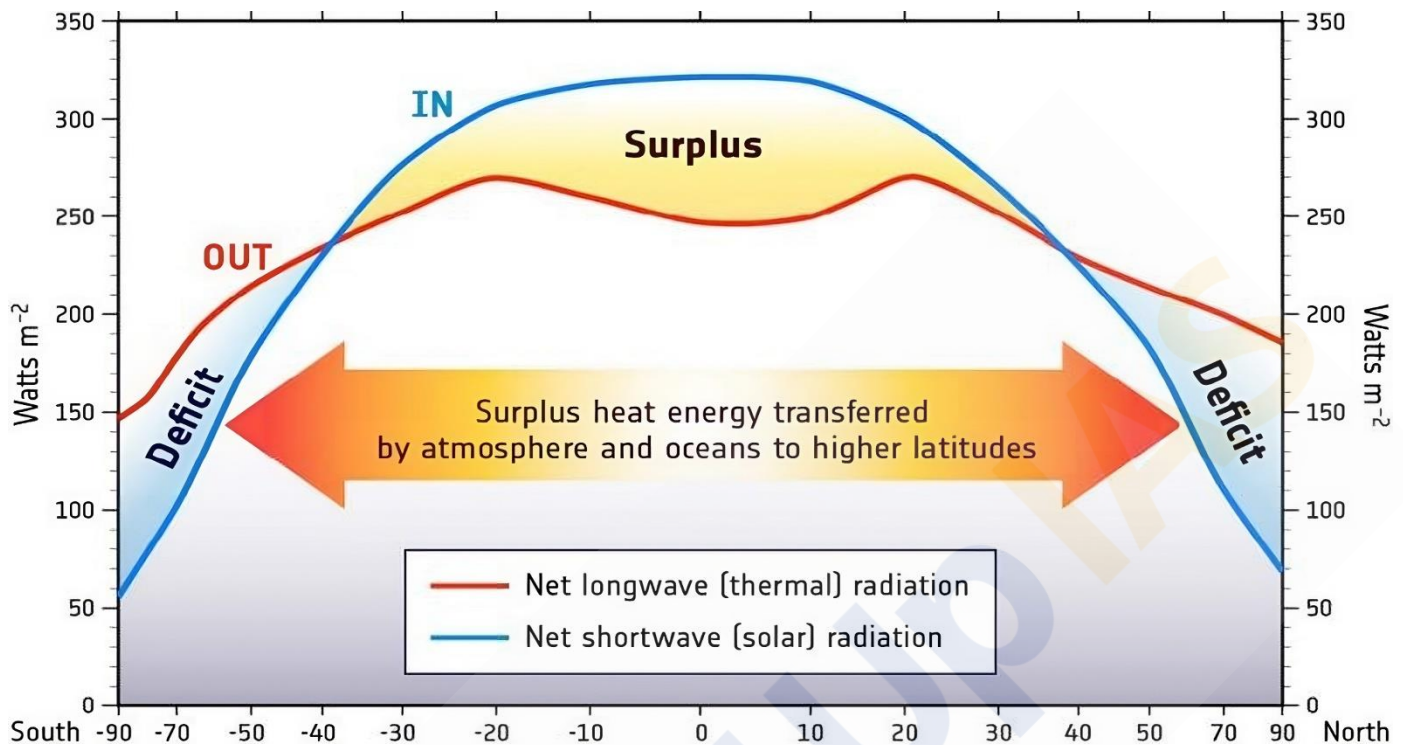
Mechanisms of Heat Redistribution:

The imbalance created by insolation at different latitudes is partially **mitigated by atmospheric circulation, winds, and ocean currents**. These mechanisms play a crucial role in **transferring heat from surplus heat regions to deficit heat regions**, thereby maintaining a latitudinal heat balance.

- Atmospheric Circulation:** The **movement of air masses** in the atmosphere redistributes heat from the equator towards the poles through convection and atmospheric dynamics.
- Ocean Currents:** Similarly, **ocean currents** act as conveyor belts of heat, transferring warm water from tropical regions towards higher latitudes and cold water from polar regions towards the equator.

Importance of Latitudinal Heat Balance:

The latitudinal heat balance is essential for **regulating global climate patterns and maintaining overall climate stability**. By ensuring that excess heat from the tropics is dispersed towards polar regions, and vice versa, these mechanisms help moderate temperature extremes and influence regional weather patterns worldwide.



Temperature and Its Distribution:

Temperature indicates the **relative degree of heat of a substance**. Heat is the energy that makes objects hot, while **temperature measures the intensity of this heat**. Although distinct from each other, heat and temperature are closely related because gaining or losing heat is necessary to raise or lower the temperature.

Distribution of temperature varies both **horizontally and vertically**. This can be studied under two categories:

- **Horizontal Distribution of Temperature**
- **Vertical Distribution of Temperature**

Horizontal Distribution of Temperature:

The horizontal distribution of temperature across Earth's surface is depicted using "**isotherms**," which are **lines connecting points with equal temperatures**. The term "isotherm" combines 'iso' (equal) and 'therm' (temperature). If you study an isotherm map you will find that the **distribution of temperature is uneven due to various factors**.

Factors responsible for the distribution of temperature:

Latitude:

- The angle of incidence of solar rays decreases from the equator towards the poles. **Higher angles of incidence result in higher temperatures**, while **lower angles lead to lower temperatures**. Consequently, tropical regions experience higher temperatures, which decrease significantly towards the poles, where temperatures remain below freezing for most of the year.

Land and Sea Contrast:

- **Land heats up and cools down more rapidly than water.** During the day, temperatures are higher on land, whereas at night, water retains more heat. Seasonally, air temperatures above land are higher in summer, while in winter, oceans exhibit higher temperatures compared to landmasses. Snow-covered land warms slowly due to high reflection of solar energy, while vegetation-covered land moderates' temperature by using insolation for water evaporation.

Relief and Altitude:

- Relief features like mountains, plateaus, and plains modify temperature distribution. **Mountains act as barriers to wind movement;** for example, the **Himalayas** block cold Central Asian winds, making Kolkata warmer than Guangzhou in winter despite being on similar latitudes.
 - Temperature decreases by approximately 6°C per 1000 meters of altitude, known as the normal lapse rate, making higher elevations cooler than plains. The rate of temperature decrease with altitude varies by time of day, season, and location. For instance, Quito (2800 meters above sea level) has an annual mean temperature of 13.3°C, while Guayaquil (12 meters above sea level) has 25.5°C.

Air Mass and Ocean Currents:

- **Areas influenced by warm air masses experience higher temperatures, while those under cold air masses have lower temperatures.**
 - Similarly, **warm ocean currents**, like the **North Atlantic Drift**, warm the coasts they flow along, preventing freezing in places like Northwestern Europe. Conversely, **cold currents**, such as the **Labrador Current**, cool the coasts, resulting in regions like Quebec freezing in winter despite being at lower latitudes compared to Northwestern Europe.

Winds:

- **The wind is defined as the movement of air mass from an area of high pressure to an area of low pressure.** They transport heat from one region to the other, affecting temperature distribution through the process of advection.

Vegetation Cover:

- Vegetation cover impacts soil temperature. **Soils without vegetation heat up quickly, while vegetation-covered soils absorb and retain heat, preventing rapid radiation.** Consequently, temperature variations are smaller in dense forested areas compared to deserts. For example, the annual temperature range in equatorial regions is about 5°C, while in hot deserts, it reaches up to 38°C.

Nature of the Soil:

- The **color, texture, and structure of soils influence temperature.** Black, yellow, and clayey soils absorb more heat compared to sandy soils, which radiate heat quickly. Therefore, temperature contrasts are less pronounced in areas with black soil compared to sandy soil regions.

Slope and Aspect:

- The **angle and direction of slopes control insolation receipt.** Gentle slopes receive greater solar energy concentration, raising temperatures, while steeper slopes receive less, lowering temperatures. **Southern slopes of the Himalayas are warmer due to higher solar energy concentration** compared to the colder northern slopes, which face away from the sun and are exposed to cold winds. Consequently, settlements and cultivation are more prevalent on the southern slopes, while the northern slopes remain forested.

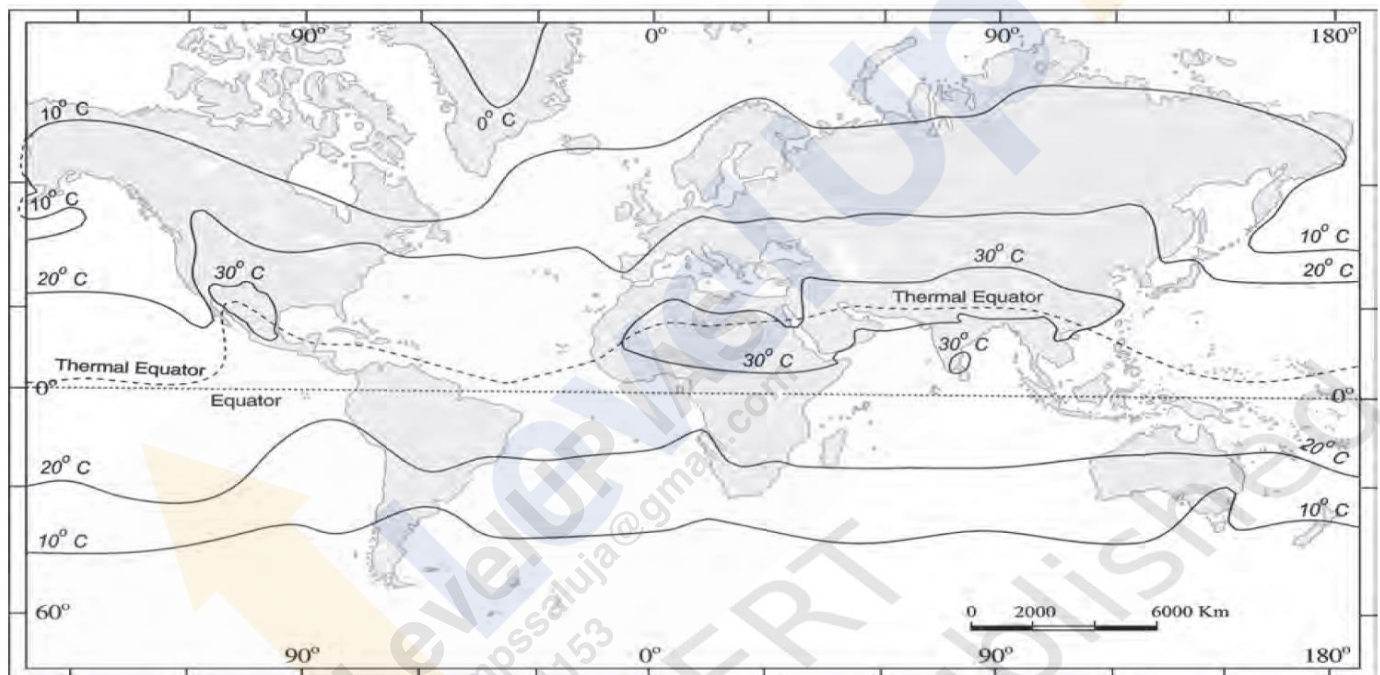
Horizontal Distribution of Temperature in January:

In January, the sun shines directly overhead near the Tropic of Capricorn, resulting in **summer** in the **southern hemisphere** and **winter** in the **northern hemisphere**. High temperatures are found over landmasses in three regions of the **southern hemisphere** - **North-west Argentina, East and Central Africa, and Central Australia**. These regions are enclosed by the **30°C isotherm**. In contrast, the **northern hemisphere's landmasses are cooler** than the oceans during this time, with North-east Asia experiencing the lowest temperatures.

Isotherm Patterns:

- **Northern Hemisphere** - The air is warmer over the oceans than over the landmasses. Consequently, **isotherms bend towards the north over the oceans and towards the south over the continent**.
- **Southern Hemisphere** - In contrast, **isotherms bend towards the south over the continent and towards the north over the oceans**.

The **southern hemisphere has a large expanse of water**, resulting in **regular and widely spaced isotherms**. In contrast, the **northern hemisphere, with its large landmasses, has irregular and closely spaced isotherms**. This difference leads to less extreme seasonal contrasts between land and water in the southern hemisphere's middle and higher latitudes compared to those north of the equator.



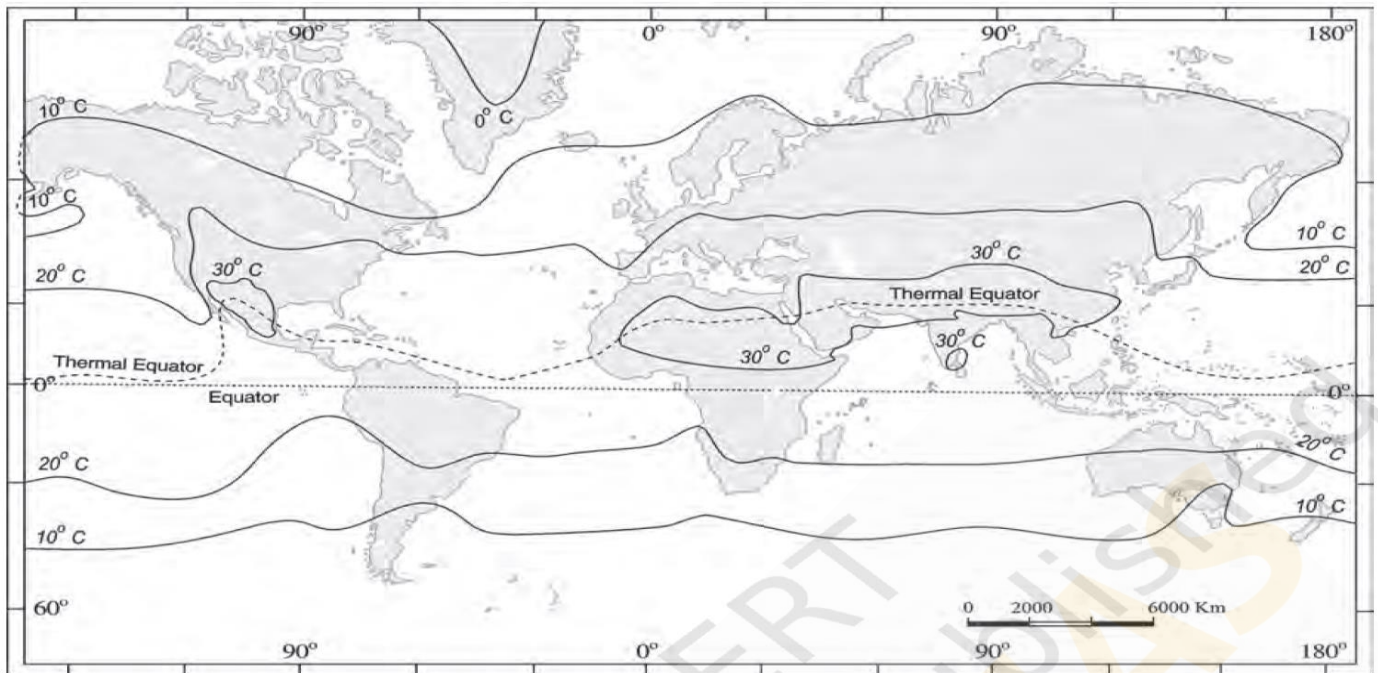
Horizontal Distribution of Temperature in July:

(see it)

During July, the sun is positioned vertically overhead near the Tropic of Cancer. It is **summer** in the **northern hemisphere** and **winter** in the **southern hemisphere**. The isothermal behavior is the opposite of what it is in January. **Isotherm of 30°C passes between 10° N and 40° N latitudes**. The regions having this temperature include South Western USA, the Sahara, the Arabia, Iraq, Iran, Afghanistan, desert region of India and China.

Isotherm Patterns:

- **Northern Hemisphere** – Isotherms **bend towards the north over the continents**, indicating overheating of landmasses. Hot tropical winds penetrate deeply into northern interiors. **Over oceans, isotherms bend towards the south** (equatorward), suggesting cooler oceanic temperatures moderate tropical interiors.
- **Southern Hemisphere** - Isotherms are wide spaced over oceans while they are closely spaced over landmasses. It **generally run parallel to the latitude, moving south over continents and north (equatorward) over oceans**.



Comparing July and January isotherm maps reveals significant patterns. The highest temperatures migrate latitudinally with the movement of the sun's vertical rays. **In both hemispheres, the highest temperatures occur in low latitudes, gradually decreasing towards the poles due to the decreasing solar insolation.**

In the northern hemisphere, isotherms departing from landmasses sharply bend towards the poles in winter and towards the equator in summer. This behavior reflects the **differential heating and cooling rates of continents versus oceans**, with **landmasses experiencing more extreme temperature fluctuations** throughout the year.

Vertical Distribution of Temperature:

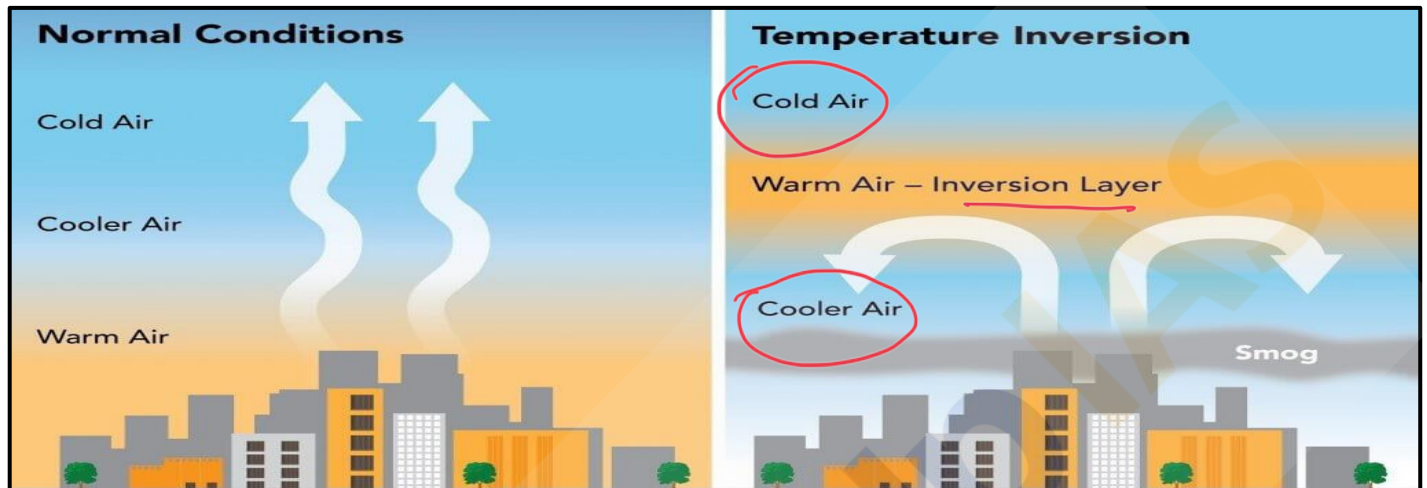
Temperature decreases with increasing height in the troposphere but the rate of decrease varies according to **seasons, duration of sunshine and location**. On average, the rate of decrease of temperature with increasing altitudes in a stationary column of air with the absence of any vertical motion is **6.5 °C per 1000 meters**. This decrease in temperature is called a **vertical temperature gradient or normal Lapse rate**. It should be mentioned that the normal lapse rate is average but the **actual lapse rate varies both spatially and temporally**. The actual lapse rate is called **environmental lapse rate**.

- **Troposphere:** The troposphere shows a **decrease in temperature with height**, ranging from 16 km over the equator to 6 km over the poles. This height varies seasonally, being higher in summer than in winter. The upper limit is called the tropopause.
- **Stratosphere:** Above the tropopause, in the stratosphere, the **temperature increases with height**. It reaches 0°C or 32°F at the stratopause, which is the upper limit of the stratosphere at a height of 50 km from sea level. This increase in temperature is due to the absorption of radiation by ozone.
- **Mesosphere:** In the mesosphere, which extends from 50 km to 80 km, the **temperature again decreases with increasing height**. At the mesopause, the upper limit of the mesosphere, the temperature drops to -80°C.
- **Thermosphere:** Beyond the mesopause, in the thermosphere, the **temperature increases with height**. It is estimated that the temperature at the upper limit of the thermosphere (height undecided) reaches 1700°C. However, this temperature cannot be measured by an ordinary thermometer due to the very low air density, causing the gases to become very light.

Temperature Inversion or Thermal inversion:

Normally, as we ascend through the troposphere from the Earth's surface, the temperature decreases with altitude. However, due to certain local conditions, the **temperature may increase with height instead**. This phenomenon is known as **temperature inversion** (or thermal inversion) and is also referred to as a negative lapse rate.

Temperature inversion represents a reversal of the typical temperature pattern in the troposphere. In this meteorological phenomenon, **a layer of warm air lies above a layer of cold air**. It can be caused by stagnant atmospheric conditions or by the horizontal or vertical movement of air.



Ideal Conditions for Temperature Inversion: { what u want is more reflection }

- Long winter nights: Loss of heat by terrestrial radiation from the ground surface during night may exceed the amount of incoming solar radiation.
- Cloudless and clear sky: Loss of heat through terrestrial radiation proceeds more rapidly without any obstruction.
- Dry air near the ground surface: It limits the absorption of the radiated heat from the Earth's surface.
- Slow movement of air: It results in no transfer or mixing of heat in the lower layers of the atmosphere.
- Snow covered ground surface: It results in maximum loss of heat through reflection of incoming solar radiation.

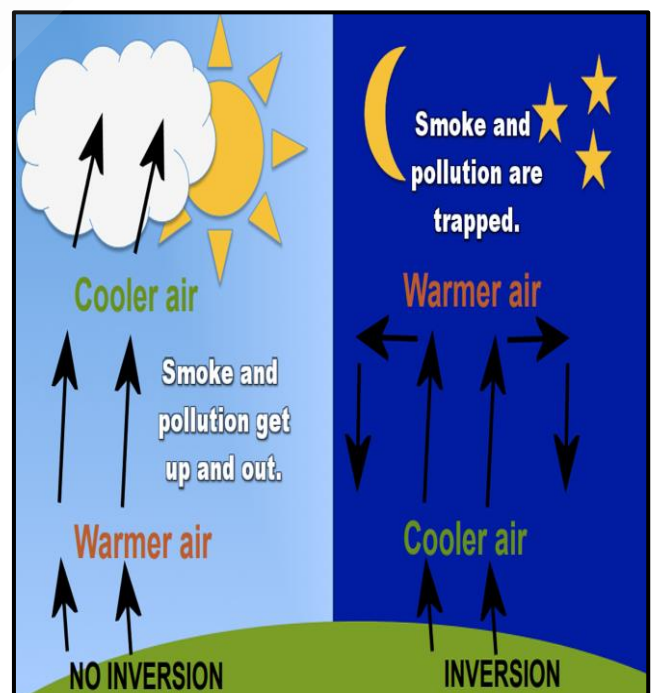
Types of Temperature Inversion:

Surface Temperature Inversion (Ground Inversion):

A ground inversion forms when air is cooled by contact with a colder surface, becoming cooler than the overlying atmosphere. This phenomenon occurs most frequently on **clear nights** when the ground rapidly cools by radiation. If the surface air temperature drops below its dew point, fog may form.

Ground inversions are **common in higher latitudes**. In **lower and middle latitudes**, surface temperature inversions occur during cold nights and dissipate during the daytime, typically disappearing with sunrise.

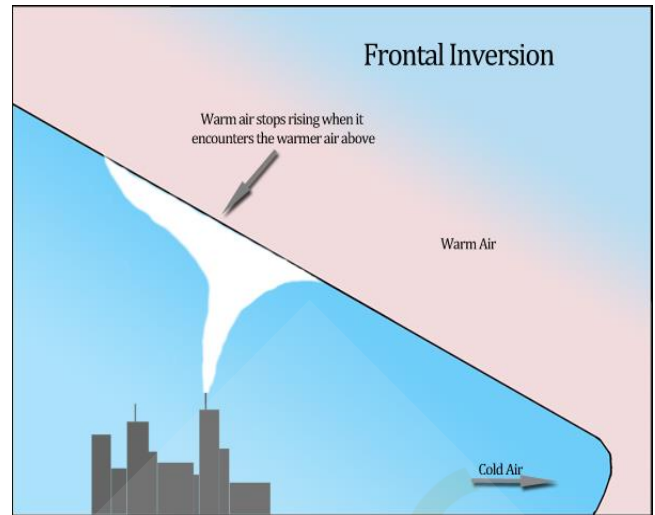
The duration and height of surface inversions increase towards the poles. The conditions required for ground surface inversion include **long winter nights**, **cloudless calm skies**, **dry air with low relative humidity**, **a calm atmosphere** or slow air movement, and a snow-covered surface.



Frontal Inversion:

Frontal inversion occurs due to the **horizontal and vertical movement of air masses**. It occurs when a **cold air mass displaces a warm air mass** (commonly known as Cold and Warm Fronts), and the cold air mass lifts the warm air mass aloft. This configuration reverses the normal lapse rate, leading to a temperature inversion.

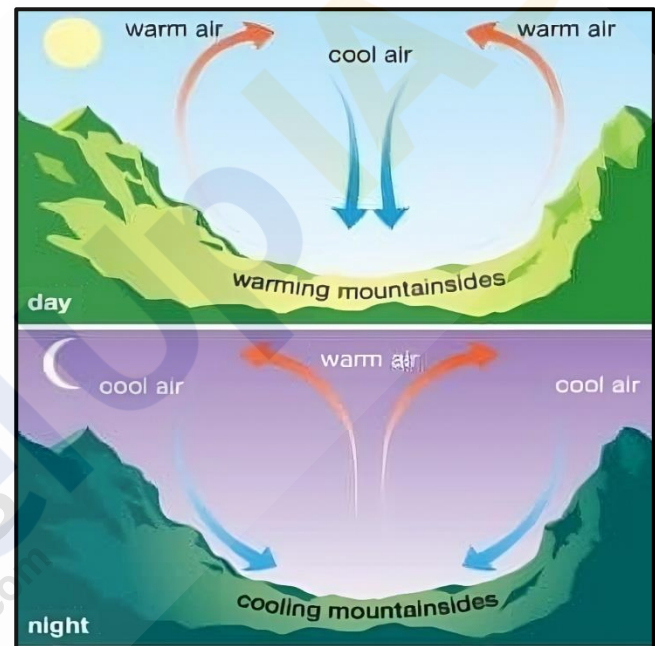
Unlike other inversions, which are typically horizontal, frontal inversions have a noticeable slope. Additionally, the **humidity is often high, and clouds may form just above the inversion layer**. This type of inversion is unstable and dissipates as weather conditions change.



Intermontane Valley Inversion:

Intermontane valley inversion generally occurs in **mountainous valleys** due to radiation and the vertical movement of air, also known as **vertical advective inversion of temperature**.

During **winter nights**, the temperature in the **upper parts of valleys** in mountainous regions becomes **excessively cold because of the rapid loss of heat from the surface** through terrestrial radiation. As a result, the air in contact with the cool surface also becomes cool. Conversely, the valley floor's temperature does not fall significantly due to the comparatively lower rate of heat loss through terrestrial radiation. Consequently, the **air on the valley floor remains warmer than the air above**.



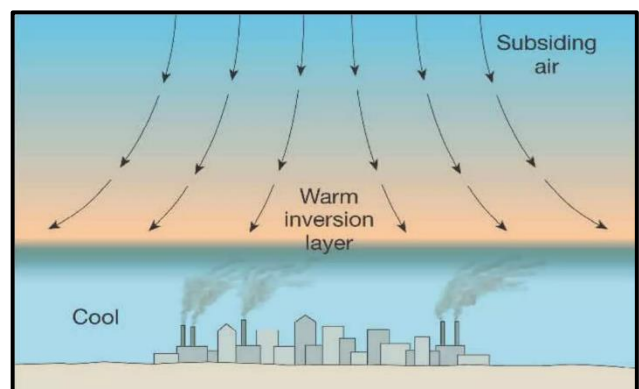
The warm, light air on the valley floor is pushed upward by the descending cold, heavier air from the upper part of the valley. This results in **warm air aloft and cold air at the valley floor, causing a temperature inversion**.

This condition leads to severe **frost on the valley floors, causing significant damage to fruit orchards, vegetables, and agricultural crops, while the upper parts of the valleys remain free from frost**. This is why **human settlements in mountainous valleys of middle latitudes are typically located on the upper parts rather than the valley floors**.

Subsidence Inversion (Upper Surface Temperature Inversion):

When a **widespread layer of air descends, it is compressed and heated by the resulting increase in atmospheric pressure**, and as a result the lapse rate of temperature is reduced. The air at higher altitudes **becomes warmer than at lower altitudes, producing a temperature inversion**. This type of temperature inversion is called **subsidence inversion**.

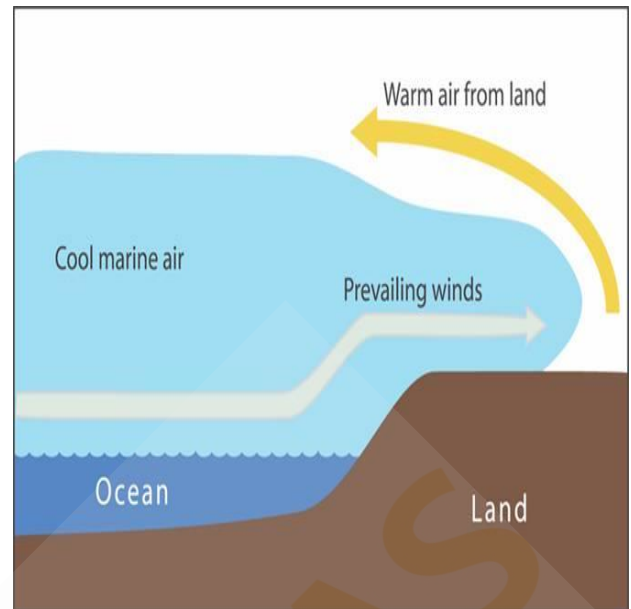
It is very common over the northern continents in winter (dry atmosphere) and over the subtropical oceans. These regions generally have subsiding air because they are located under large high-pressure centers. It is also called upper surface temperature inversion because it takes place in the upper parts of the atmosphere.



Marine Inversion:

Marine inversions **occur near large bodies of water**, particularly during spring when the water is coldest. As **air flows over these bodies of water, it cools as heat is transferred from the air to the colder water**. This cooled air, denser than the warmer air over land, **moves inland beneath the warmer air layer, creating an inversion**.

This phenomenon is common in coastal regions where cool, moist air from the ocean contrasts with warmer, drier air over land, leading to stable atmospheric conditions that can persist for some time. Marine inversions often result in **foggy conditions near coastal areas** and can influence local weather patterns and visibility.



Effects of Temperature Inversion:

- **Air Quality Issues:** Temperature inversions trap pollutants close to the ground. This happens because warm air above prevents the rising of cooler air and pollutants, leading to higher concentrations of pollutants (like smog, particulate matter, and ozone) in the lower atmosphere. This can worsen respiratory conditions and reduce visibility.
- **Weather Conditions:** Inversions can lead to stable atmospheric conditions where clouds and precipitation may be inhibited. This stability can contribute to the formation of fog and low-level clouds, impacting transportation and aviation.
- **Temperature Distribution:** Normally, temperature decreases with altitude in the troposphere. During an inversion, the opposite occurs, leading to warmer air aloft and colder air near the surface. This can affect regional weather patterns and may contribute to unusual temperature distributions.
- **Impact on Agriculture:** Winter crops such as wheat, barley, mustard, vegetables, chilies, and potatoes suffer significant damage. In the northern plains of India, particularly in UP, Punjab, and Haryana, sugarcane crops develop diseases such as red rot, which diminishes their sugar content.
- **Radiation Effects:** Inversions can affect the radiation balance at the Earth's surface by trapping outgoing longwave radiation during nighttime inversions, leading to cooler nighttime temperatures.
- **Aviation and Transportation:** Inversions can create hazardous conditions for aviation due to reduced visibility and the potential for fog formation. They can also affect road transportation by reducing visibility and causing road surface icing.
- **Thunderstorms and tornadoes:** Intense thunderstorms and tornadoes are also associated with inversion of temperature because of the intense energy that is released after an inversion blocks an area's normal convection patterns.